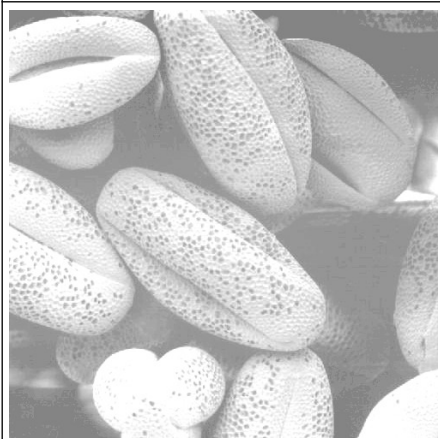
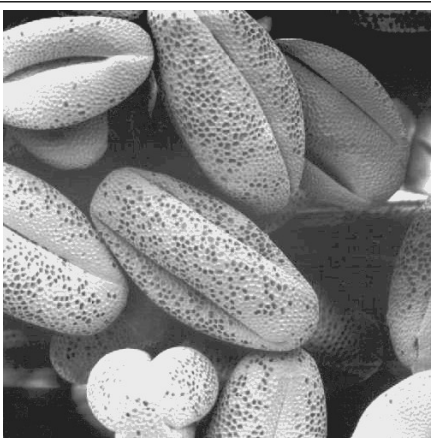
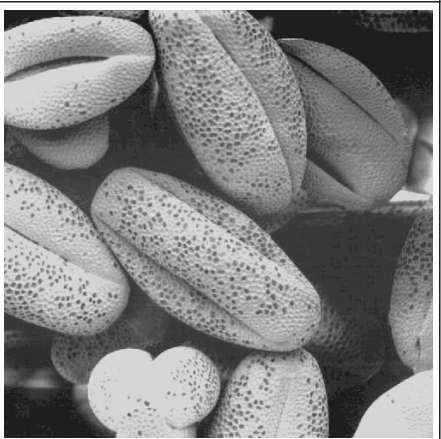


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September 11, 2026

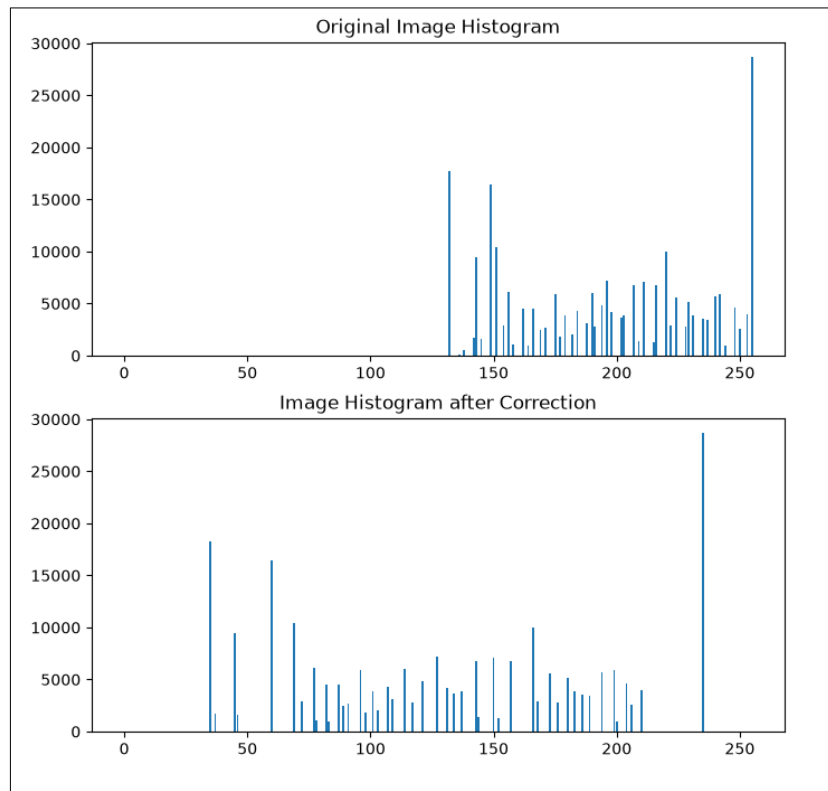
Assignment One

Source image: LowContrast-a

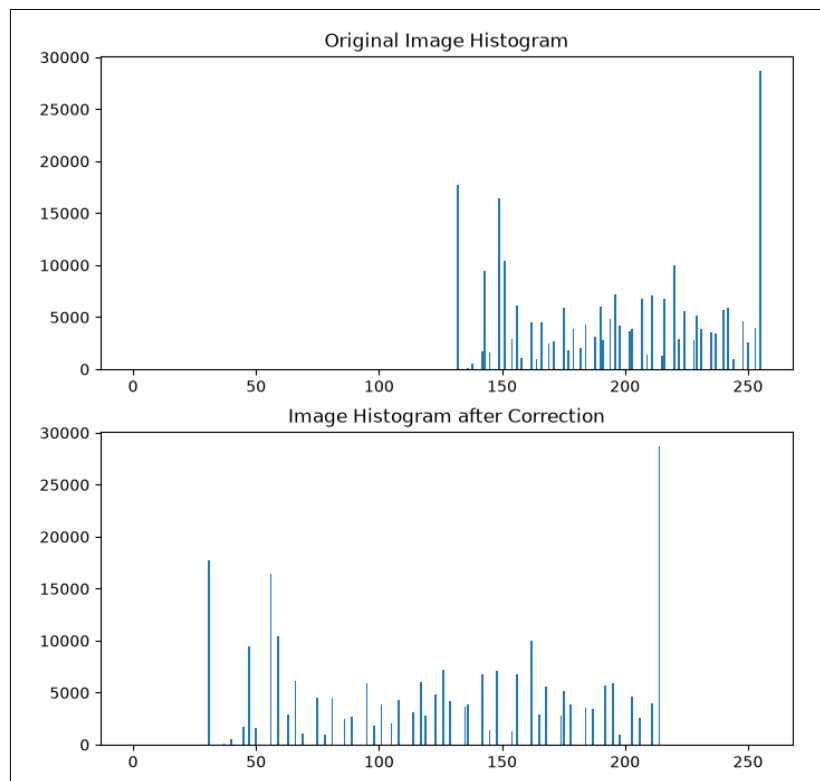
	Original Image	Image after Histogram Equalization (min=30, max=235)	Image after Linear Contrast Correction (mean=127, std=60)
Min	132	35	31
Max	255	235	214
Mean	196.35	131.14	126.55
Variance	1620.82	3944.16	3598.8
Standard Deviation	40.26	62.80	59.99
Signal-to-noise ratio	4.88	2.09	2.11

Original	After Histogram Equalization	After Linear Contrast Correction
		

Histogram of Original Image vs. Image after Histogram Equalization:

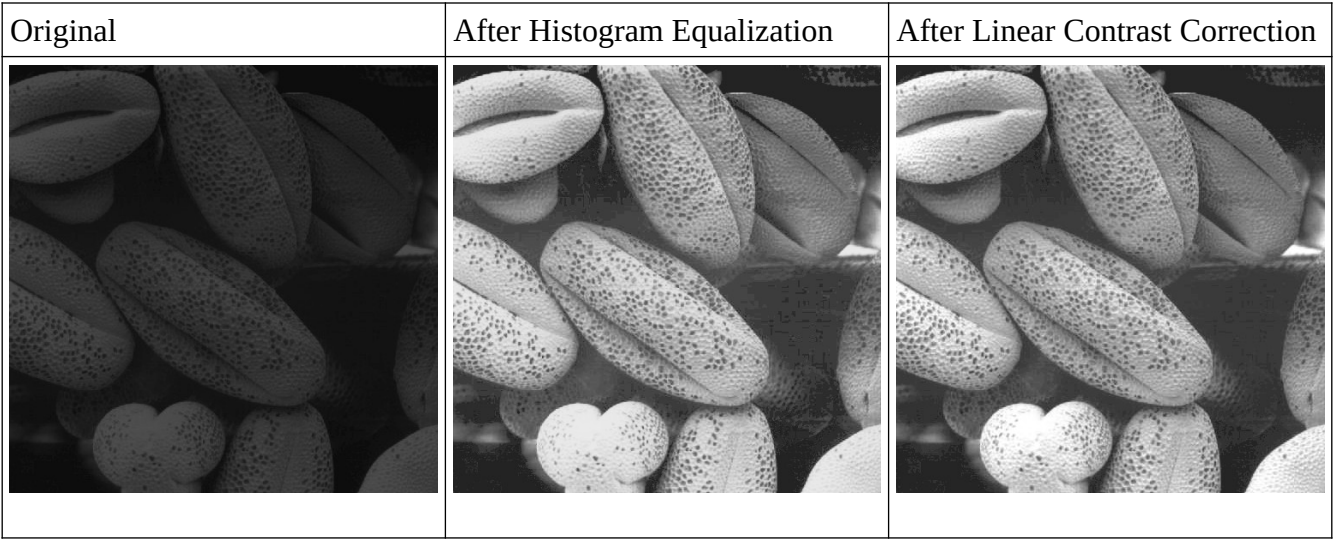


Histogram of Original Image vs. Image after Linear Contrast Correction:

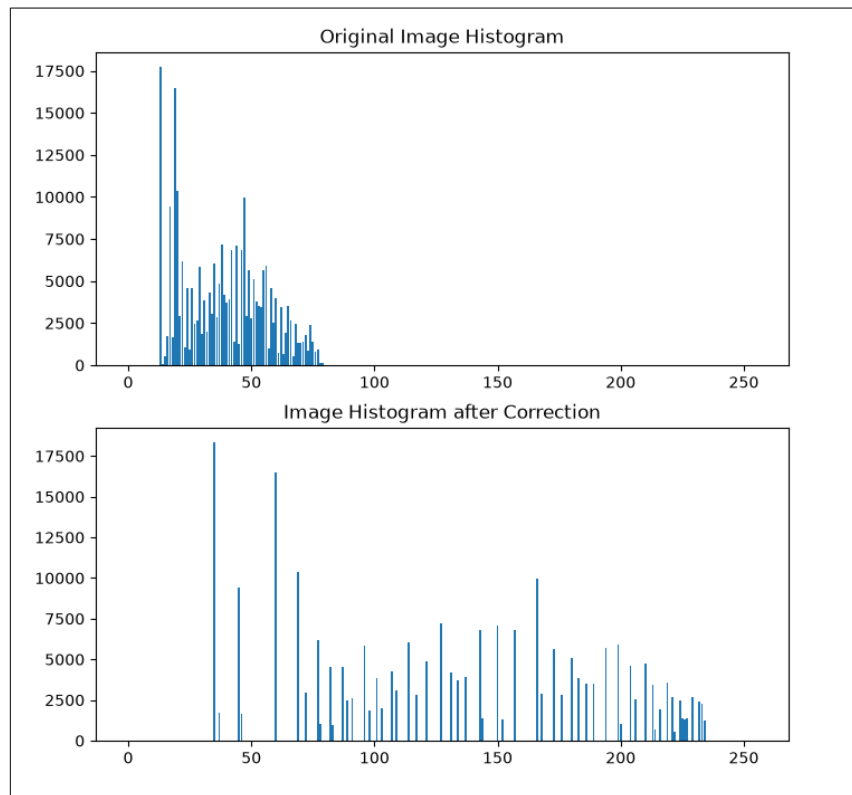


Source image: LowContrast-b

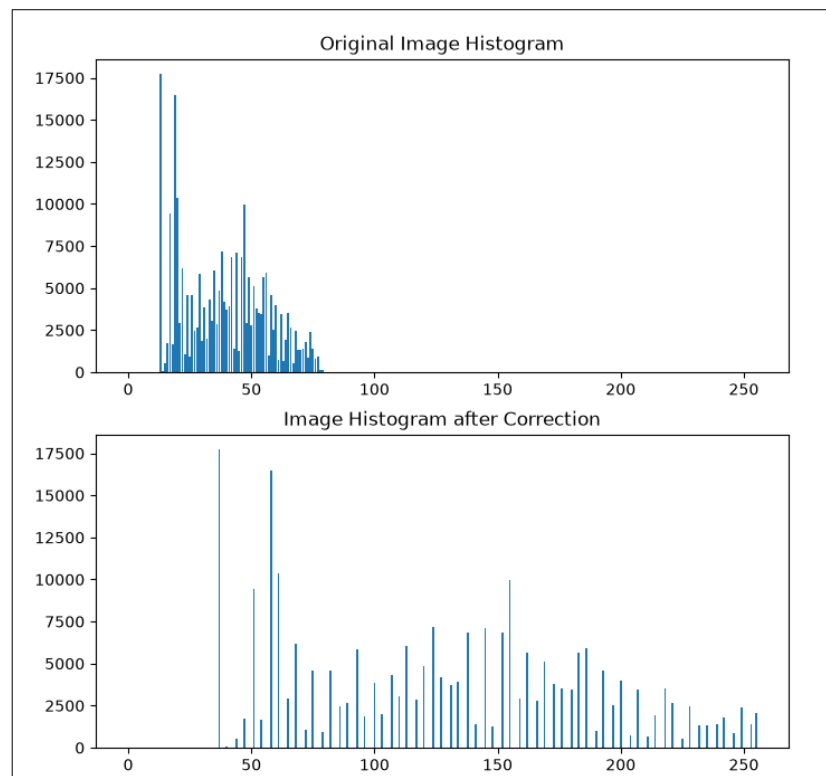
	Original Image	Image after Histogram Equalization (min=30, max=235)	Image after Linear Contrast Correction (mean=127, std=60)
Min	13	35	37
Max	83	235	255
Mean	38.75	129.78	126.45
Variance	297.8	3681.79	3589.87
Standard Deviation	17.27	60.68	59.92
Signal-to-noise ratio	2.25	2.13	2.11



Histogram of Original Image vs. Image after Histogram Equalization:



Histogram of Original Image vs. Image after Linear Contrast Correction:



Conclusions:

The two methods of contrast correction are effective ways to improve the contrast of images whose contrast is too high or too low. Both histogram equalization and linear contrast correction produced similar resulting images with similar statistics and produced images with a more balanced histogram. However, I noticed that both methods created some artifacts in the image after correction. For example, LowContrast-a's histogram leans to the right and after correction the image contains visual noise in areas of the image which originally had darker intensities.